RATING METHODOLOGY

12 September 2025

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Contacts

Christian Hendker, CFA +49.69.70730.735
Associate Managing Director
christian.hendker@moodys.com

Kathrin Heitmann +1.212.553.4694
Associate Managing Director
kathrin.heitmann@moodys.com

Matthew Moore +61.2.9270.8108
Associate Managing Director
matthew.moore@moodys.com

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Rating Methodology

Building Materials

This rating methodology replaces the *Building Materials* methodology published on 10 September 2021. We have made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. These updates do not change our methodological approach.

Scope

This rating methodology applies to companies globally that are primarily engaged in the production, sale and distribution of building materials, such as cement, concrete, aggregates, gypsum, bricks and roof tiles.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

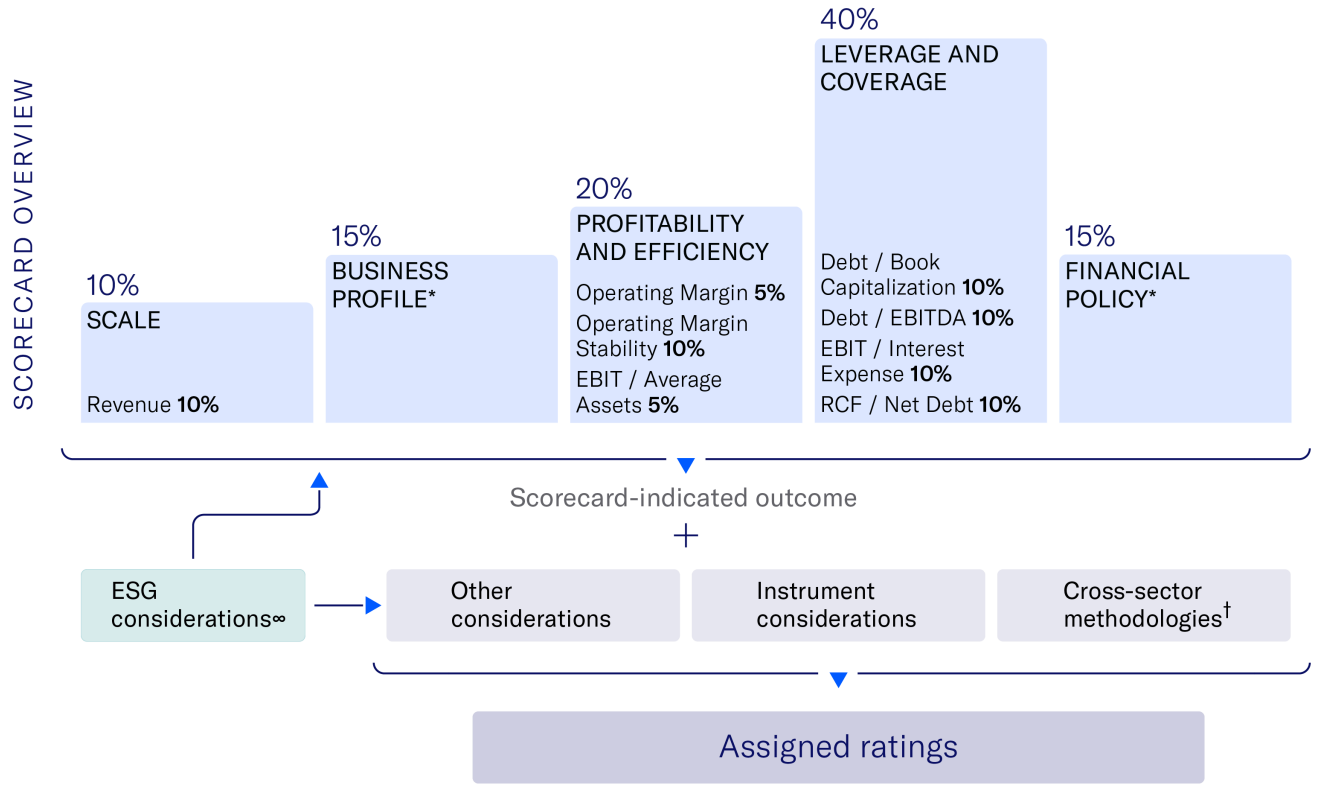
We rate companies that are primarily engaged in the construction or refurbishment of infrastructure or buildings for commercial or public use using our methodology for construction companies.[1] We rate companies that are primarily engaged in the construction and sale of finished single- and multifamily housing and large-scale residential apartments using our methodology for homebuilding and property development companies.

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the building materials rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Building materials scorecard

	SCALE (10%)	BUSINESS PROFILE (15%)	PROFITABILITY and EFFICIENCY (20%)		LEVERAGE and COVERAGE (40%)					FINANCIAL POLICY (15%)
	Revenue (USD Billion)[1] (10%)	Business Profile (15%)	Operating Margin[2] (5%)	Operating Margin Stability (10%)	EBIT / Average Assets[3] (5%)	Debt / Book Capital-ization[4] (10%)	Debt / EBITDA[5] (10%)	EBIT / Interest Expense[6] (10%)	RCF / Net Debt[7] (10%)	Financial Policy (15%)
Aaa	≥ $50	Exceptional geographic diversity in emerging and developed countries on several continents; exclusive provider of extremely diverse group of products; extremely high barriers to entry in essentially all products; extremely high degree of vertical integration; and essentially no risk from carbon regulation and a potential competitive advantage.	≥ 40%	Essentially no volatility of operating margin throughout cycles; essentially no revenue volatility; extremely diverse mix of residential, commercial and infrastructure end-markets; extremely high share of revenue from renovation; extremely strong record of quickly passing on price increases to customers; and extremely low-cost operations with very significant structural cost advantages, and a highly variable cost base.	≥ 25%	≤ 20%	≤ 0.5x	≥ 20x	≥ 70%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.
Aa	$30 - $50	Excellent geographic diversity in emerging and developed countries on several continents; highly diverse group of products and No. 1 or No. 2 market positions in majority of products; very high barriers to entry in most product; very high degree of vertical integration; essentially no risk from carbon regulation.	30% - 40%	Very low volatility of operating margin throughout cycles; very low revenue volatility; highly diverse mix of residential, commercial and infrastructure end-markets; very high share of revenue from renovation; very strong record of quickly passing on price increases to customers; and very low-cost operations with significant structural cost advantages and a highly variable cost base.	15% - 25%	20% - 30%	0.5x - 1x	15x - 20x	50% - 70%	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.
A	$15 - $30	Very good geographic diversity in emerging and developed countries on at least two continents; well-diversified group of products and No. 1, No. 2 or No. 3 market positions in majority of products; high barriers to entry in most products; high degree of vertical integration; very limited risk from carbon regulation.	20% - 30%	Low volatility of operating margin throughout cycles; low revenue volatility; well-diversified mix of residential, commercial and infrastructure end-markets; high share of revenue from renovation; strong record of quickly passing on price increases to customers; moderately low-cost operations with structural cost advantages and a variable cost base.	10% - 15%	30% - 40%	1x - 2x	7x - 15x	35% - 50%	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.

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	Revenue (USD Billion)[1] (10%)	Business Profile (15%)	Operating Margin[2] (5%)	Operating Margin Stability (10%)	EBIT / Average Assets[3] (5%)	Debt / Book Capital-ization[4] (10%)	Debt / EBITDA[5] (10%)	EBIT / Interest Expense[6] (10%)	RCF / Net Debt[7] (10%)	Financial Policy (15%)
Baa	$5 - $15	Good geographic diversity in emerging and developed countries on at least two continents; diverse group of products, with moderately strong market positions in majority of products; high barriers to entry in key products; somewhat high degree of vertical integration; limited carbon regulation risk.	15% - 20%	Moderate volatility of operating margin throughout cycles; moderate revenue volatility; diverse mix of residential, commercial and infrastructure end-markets; moderately high share of revenue from renovation; moderately strong record of quickly passing on price increases to customers; low-cost operations with some structural cost advantages and some variability in cost base.	7.5% - 10%	40% - 50%	2x - 3.5x	4.5x - 7x	20% - 35%	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.
Ba	$1.5 - $5	Moderate geographic diversity; moderately diverse group of products, with moderate market positions; moderately high barriers to entry in some products; moderate vertical integration; moderate carbon regulation risk.	10% - 15%	Somewhat elevated volatility of operating margin throughout cycles; somewhat elevated revenue volatility; business mix is fairly concentrated in residential and commercial end-markets with low exposure to infrastructure; moderately low share of revenue from renovation; moderately strong record of passing on price increases to customers, but with some time lag; moderately high-cost operations with no meaningful cost advantages but some variability in cost base.	4% - 7.5%	50% - 70%	3.5x - 4.5x	3x - 4.5x	10% - 20%	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.
B	$0.5 - $1.5	Low geographic diversity; product concentration or weak market positions; low barriers to entry in most products; limited vertical integration; high carbon regulation risk.	5% - 10%	Substantial volatility of operating margin throughout cycles; substantial revenue volatility; business mix is very concentrated in residential and commercial end-markets with very low exposure to infrastructure; low share of revenue from renovation; weak record of passing on price increases to customers and with a time lag; high-cost operations with limited variability in cost base.	2% - 4%	70% - 80%	4.5x - 6x	1x - 3x	5% - 10%	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.

	SCALE (10%)	BUSINESS PROFILE (15%)	PROFITABILITY and EFFICIENCY (20%)		LEVERAGE and COVERAGE (40%)					FINANCIAL POLICY (15%)
	Revenue (USD Billion)[1] (10%)	Business Profile (15%)	Operating Margin[2] (5%)	Operating Margin Stability (10%)	EBIT / Average Assets[3] (5%)	Debt / Book Capital-ization[4] (10%)	Debt / EBITDA[5] (10%)	EBIT / Interest Expense[6] (10%)	RCF / Net Debt[7] (10%)	Financial Policy (15%)
Caa	$0.25 - $0.5	Regional or niche provider with high product concentration or very weak market positions; very low barriers to entry in most products; minimal vertical integration; or very high carbon regulation risk.	2.5% - 5%	High volatility of operating margin throughout cycles; high revenue volatility; business mix is highly concentrated in residential and commercial end-markets with extremely low exposure to infrastructure; or extremely low share of revenue from renovation; very weak record of passing on price increases to customers; or very high-cost operations with very limited variability in cost base.	1% - 2%	80% - 90%	6x - 7x	0.5x - 1x	2.5% - 5%	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.
Ca	< $0.25	Regional or niche provider with very high product concentration or extremely weak and eroding market positions; essentially no barriers to entry in essentially all product lines; essentially no vertical integration; or extremely high carbon regulation risk.	< 2.5%	Extremely high volatility of operating margin throughout cycles; extremely high revenue volatility; business mix is concentrated in residential and commercial end-markets with no exposure to infrastructure; or no revenue from renovation; extremely weak record of quickly passing on price increases to customers; or extremely high-cost operations with extremely limited variability in cost base.	< 1%	> 90%	> 7x	< 0.5x	< 2.5%	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 60%. A value of 60% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 40%. A value of 40% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is zero. A value of 0% equates to a numeric score of 0.5. The Ca endpoint value is 110%. A value of 110% or worse equates to a numeric score of 20.5, as does a negative Debt/Book Capitalization value.
[5] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 9x. A value of 9x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[6] For the linear scoring scale, the Aaa endpoint value is 30x. A value of 30x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5, as does negative EBIT.
[7] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 90%. A value of 90% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (10% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

Large-scale companies generally have more flexibility to allocate capacity and absorb expenses under different demand and cost scenarios than small-scale companies, an important consideration in the highly cyclical building materials industry.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (15% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Core aspects of a building materials company's business profile include its geographic diversity, market position, product diversity, barriers to entry, degree of vertical integration and exposure to carbon regulation.

Geographic diversity is an important credit strength because of the industry's vulnerability to economic downturns. The fundamental business drivers affecting building materials companies are generally correlated with gross domestic product (GDP) growth and construction activity. A company with greater geographic diversity can typically better withstand local or regional economic weakness than one that is geographically concentrated. Exposure to emerging markets can improve a company's growth prospects because the consumption of building materials as a percentage of GDP typically rises with growth in GDP per capita. However, emerging markets generally exhibit greater volatility in economic growth than developed markets, underscoring the importance of geographic diversity.

A company's market position is a meaningful indicator of its resilience to economic downturns and intensifying competition. Companies with leading market positions are generally more likely to pass on increases in input costs to customers. Although few

viable alternatives to the sector's core products exist at similar price points, which diminishes secular pressure and technology risk, product diversity is important because it helps companies manage revenue weakness or rising costs, whether due to competitive or cyclical conditions, in any one product line.

Barriers to entry, which can vary by region and product line, are also critical because they help strongly established companies defend their market positions, and high entry barriers make it difficult for new entrants to gain share.

Vertical integration is another important aspect of a company's business profile. A company that produces its own raw materials, converts the materials into finished goods and distributes those goods is less reliant on other companies and therefore more in control of its cost base.

Carbon regulation is an important consideration because it could increase costs for the energy-intensive building materials sector if companies cannot pass on compliance costs to customers or reduce their carbon emissions. The impact of carbon regulation varies among companies because of disparate regulations across regions and products, and because of the different energy intensity involved with producing various building materials, again highlighting the importance of geographic and product diversification.

How we assess it for the scorecard

We score this factor based on our qualitative assessment of the durability of demand and a company's ability to meet that demand.

We generally do not expect a given company to match each attribute listed for a given scoring category for this factor. We typically assign a factor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on the factor score. For example, reliance on a single region or product line may limit the factor score to the Ba category or lower even if the company's market position is strong and its carbon regulation risk is low, because the business risk in these cases is typically high.

We assess qualitatively a company's geographic diversity, market position, product diversity, barriers to entry, degree of vertical integration and exposure to carbon regulation.

In assessing geographic diversity, we typically consider the amount of revenue a company receives from customers in each country in which it operates, as well as whether their economies are developed or emerging. Because building materials companies typically sell their products within the region of their operations due to generally high transport costs, the proportion of revenue companies receive from operations in each country relative to the assets in each country is generally fairly comparable.

We typically assess a company's market position within a given country, in light of the local nature of the business given the low value-to-weight ratio of products sold. In assessing a company's market position, we typically consider market share, which we estimate based generally on information provided by third parties and by companies. Companies with leading market positions and high market share generally score higher than those with weak market positions and low market share. We assess product diversity by considering the number of discrete product segments of a company. Examples of product segments include cement, ready-mixed concrete and aggregates.

Our assessment of barriers to entry typically considers the company's regions and product segments. We may consider the company's access to pits and quarries necessary to extract raw materials from the ground as well as its production and distribution network. Rights and permits to extract raw materials can also be a strong barrier to entry.

Different product segments generally require different levels of start-up investment. Also, the generally high transport costs and low value-to-weight ratio of many building materials create regional barriers to entry, because transport costs of heavy products rise rapidly with distance from the plant location. In the cement market, for example, the high cost of building a plant and creating efficient distribution channels reduces the likelihood of new entrants, relative to a lighter-weight product, such as ceramics. As another example, the ready-mixed concrete business requires significantly less upfront investment than the cement business, reducing barriers to entry.

Our assessment of vertical integration is related to our evaluation of product diversity in terms of product breadth, but also considers whether the company produces its own raw materials and has its own distribution channel for those products, or whether it relies on other companies for raw materials and the distribution of its products. For example, a company that produces cement, transforms it

into ready-mixed concrete, and distributes it to the end customer typically scores higher than one that produces cement but does not add value to it or distribute it, or one that relies on other companies for raw materials and for distribution of its products.

In assessing a building materials company's exposure to carbon regulation, we generally consider existing and expected regulations within a company's geographic footprint, its product segments and any steps it has taken to mitigate regulatory risk or gain a competitive advantage. For example, cement production emits more carbon than other building products, such as ceramics, so a company that derives most of its revenue from the cement market typically has greater carbon regulation risk. A company can reduce carbon emissions by investing in technology such as carbon capture storage or by shifting its product portfolio to products with lower energy intensity, thereby lowering its risk or creating competitive advantages.

Factor: Profitability and Efficiency (20% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors, Operating Margin and EBIT/Average Assets, and one qualitative subfactor, Operating Margin Stability.[3]

Operating Margin

We use the ratio of operating income to revenue. We use operating income, a measure or estimate of profitability before interest, taxes and unusual items but after depreciation costs, because reinvestment is typically essential for a company to maintain its market position.

Operating Margin Stability

We score this subfactor based on a qualitative assessment of the company's operating margins throughout varying economic conditions to assess its ability to manage revenue and costs.

We typically assess how much revenue a company receives from each of three end markets, including the residential, commercial and infrastructure end markets. Companies with a high concentration of revenue in any one end market generally receive lower scores for this subfactor than companies with diversified exposure to all three end markets. Of the three end markets, exposure to the infrastructure end market is likely to support a higher score for this subfactor than the other markets, because infrastructure spending is generally more resilient during economic downturns. Governments may invest in infrastructure projects as a contra-cyclical measure, and although spending can rise or drop based on budget approval, both projects and government budgets typically span multiple years, lending predictability once budgets are approved.

Companies that receive a high level of revenue from the renovation segment may receive higher scores for this subfactor than companies that do not, because end customers are more likely to spend on renovation projects than on new construction during economic downturns.

Another consideration for a company's revenue stability is its ability to pass on price increases to customers. We generally assess the company's cost base relative to its peers, and a company with low-cost operations, structural cost advantages and a variable cost base typically scores higher for this subfactor than a company with higher-cost operations and a highly fixed cost base. A company may derive structural cost advantages from, for example, proximity to customers or lower cost access to raw materials. Our assessment of the cost base may also consider the company's overall energy costs and its flexibility to switch between fuel sources.

EBIT / Average Assets

We use the ratio of earnings before interest and taxes (EBIT) to total assets averaged over the past two years. This ratio is an indicator of a company's ability to generate a meaningful return relative to the level of investment in its asset base. It is also a useful indicator in comparing companies that make products with different asset intensity.

Factor: Leverage and Coverage (40% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on four quantitative subfactors: Debt/Book Capitalization; Debt/EBITDA; EBIT/Interest Expense; and RCF/Net Debt.[4]

Debt / Book Capitalization

We use the ratio of total debt to book capitalization.

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Increasing environmental requirements and efforts to reduce greenhouse gas emissions (known as carbon transition risk) may lead to higher costs for building materials companies, especially cement producers. Stricter air pollution standards could also increase costs. Inability to pass compliance costs on to customers could erode profitability and cash flow generation. Disparities in regulations and associated costs are likely to favor some companies and create competitive challenges for others. Our expectation for environmental considerations may be an important aspect of our assessment of a company's business profile, and our expectation for fuel costs may play a role in our assessment of the operating margin stability subfactor score.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's business profile or operating margin stability. However, we may assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow

may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Seasonality

Seasonality is an important driver of customer demand and can cause swings in cash balances and working capital positions for companies in this sector. Higher volatility creates less room for errors in meeting customer demand or operational execution. Construction activities typically need to occur during mild weather conditions, which leads to a peak demand for building materials in summer and low levels of activity and demand in winter. Building materials companies may also use cold weather periods for extended maintenance activities, which may amplify seasonality effects. Seasonality effects vary depending on the products, so companies that rely heavily on a particular season for sales may have greater earnings and cash flow volatility.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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Author

Karen Berckmann, CFA

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